

Web Mining and Recommender Systems

Introduction

What is this class about?

In this course we will build
models that help us to
understand data in order to gain
insights and make **predictions**

Examples – Recommender Systems

Prediction: what (star-) rating will a person give to a product?
e.g. rating(julian, Pitch Black) = ?

Application: build a system to recommend products that people are interested in

103 of 115 people found the following review helpful

★★★★★ **Excellent Sci-Fi**

Pitch Black was arguably one of the most overlooked films of the early year. Although the setting of the film could seem routine to a casual viewer(space travelers stranded and bickering on a hostile planet infested with alien nasties), director David Twohy's wonderful use of color and stylistic flourishes more than makes up for any trivial complaints.

For...

[Read the full review >](#)

Published on September 12, 2000 by Eric J. Pray

Insights: how are opinions influenced by factors like time, gender, age, and location?

Examples – Social Networks

Prediction: whether two users of a social network are likely to be friends

Application: “people you may know” and friend recommendation systems

Insights: what are the features around which friendships form?



Examples – Advertising

Prediction: will I click on an advertisement?

Application: recommend relevant (or likely to be clicked on) ads to maximize revenue

query → engagement rings

Web Shopping Images Maps News More Search tools

About 39,300,000 results (0.33 seconds)

ads

- Tiffany & Co.® Engagement - Tiffany.com**
Ad www.tiffany.com/Engagement
Your Perfect Match Deserves The Best. Consult An Expert At Tiffany®
Tiffany & Co. has 122,932 followers on Google+
Rings
Diamond Consultation
Speak to an Expert
A Tiffany Diamond
- Unique Engagement Rings - HaroldStevens.com**
Ad haroldstevens.com/Engagement-Rings
Shop Harold Stevens - 2014 LUXURY Retailer of the Year Finalist
Wedding Bands - Antique & Vintage Rings - Beverley K - Single Stone Rings
525 B Street, Suite 150, San Diego, CA (619) 231-0520
- Engagement Rings - ritani.com**
Ad www.ritani.com/Engagement-Rings (888) 978-0802
4.8 ★★★★★ rating for ritani.com
1000s of unique engagement ring designs. Free FedEx & easy returns.
Free In-Store Previews - Handcrafted in New York - Lifetime Care Package
Create Your Own Ring - The Ritani® Difference - Ritani® Diamonds

Shop for engagement rings on Google Sponsored

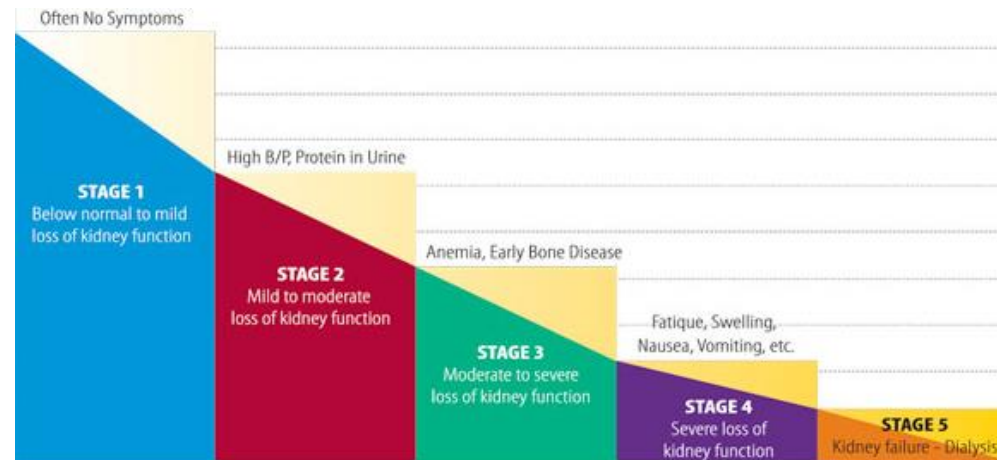
 French-Set Halo Diamond... \$1,990.00 Ritani	 18K White Gold Delicate... \$950.00 Brilliant Earth (57)	 18K White Gold Fancy D... \$1,825.00 Brilliant Earth (13)	 Chamise Diamond Eng... \$975.00 Brilliant Earth (7)
 Vintage Cushion Halo... \$4,140.00	 Princess Cut Diamond Eng... \$1,906.82	 18K White Gold Hudson... \$975.00	 18K White Gold Harmon... \$1,675.00

Insights: what products tend to be purchased together, and what do people purchase at different times of year?

Examples – Medical Informatics

Prediction: what symptom will a person exhibit on their next visit to the doctor?

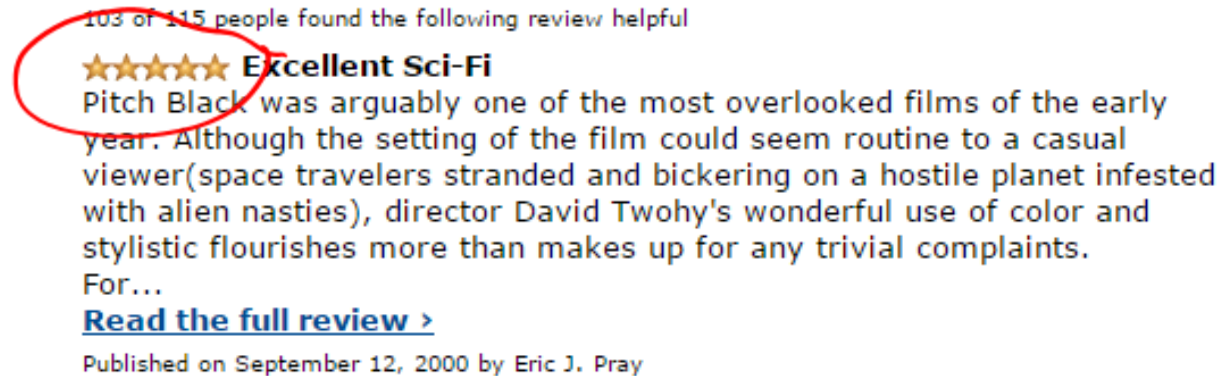
Application: recommend preventative treatment



Insights: how do diseases progress, and how do different people progress through those stages?

What we need to do data mining

1. Are the data associated with meaningful outcomes?
 - Are the data **labeled**?
 - Are the instances (relatively) independent?



e.g. who likes this movie?

Yes! "Labeled" with a rating

e.g. which reviews are sarcastic?

No! Not possible to objectively identify sarcastic reviews

What we need to do data mining

2. Is there a clear objective to be optimized?
- How will we **know** if we've modeled the data well?
 - Can actions be taken based on our findings?

103 of 115 people found the following review helpful

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For...

[Read the full review >](#)

Published on September 12, 2000 by Eric J. Pray

e.g. who likes this movie?

How wrong were our predictions on average?

$$\frac{1}{N} \sum_{\text{ratings } r_{u,i}}^N (r_{u,i} - \text{prediction}(u, i))^2$$

What we need to do data mining

3. Is there enough data?

- Are our results statistically significant?
- Can features be collected?
- Are the features useful/relevant/predictive?

What is covered in this course?

This course aims to teach

- How to **model** data in order to make **predictions** like those above
- How to **test and validate** those predictions to ensure that they are meaningful
- How to **reason about** the findings of our models

(i.e., “data mining”)

What is covered in this course?

But, with a focus on applications from **recommender systems and the web**

- **Web** datasets



- Predictive tasks concerned with human **activities, behavior, and opinions**
(i.e., **recommender systems**)

Expected knowledge (more like "things I won't teach")

Basic data processing

- Text manipulation: count instances of a word in a string, remove punctuation, etc.
- Graph analysis: represent a graph as an adjacency matrix, edge list, node-adjacency list etc.
- Process formatted data, e.g. JSON, html, CSV files etc.

Expected knowledge

Basic mathematics

- Some linear algebra
- Some optimization
- Some statistics (standard errors, normal/binomial distributions)

$$Ax = y \rightarrow x = (A^T A)^{-1} A^T y$$

$$\frac{d}{dx} (Ax - y)^2$$

Expected knowledge

All coding exercises will be done in **Python** with the help of some libraries (numpy, scipy, NLTK etc.)

Expected knowledge

Idea with "expected knowledge" is not that you know all of these things, but rather than you learn those that you don't **on your own**

See e.g. some student comments on the course:

Comment 1: *"I felt that the first four weeks of the course was slow... similar to all ML courses taught here, they review the same material on fundamentals of data science/machine learning"*

Comment 2: *"Difficult if you have not had any machine learning/data mining experience"*

Compared to other UCSD courses:

The most related classes are

- CSE 150/151 ("Introduction to Artificial Intelligence")
- CSE 250A ("Principles of Artificial Intelligence: Probabilistic Reasoning and Decision-Making")
- CSE 250B ("Machine Learning")

None of these courses are prerequisites for each other!

- This course is more "hands-on" – the focus here is on applying techniques from ML to real data and predictive tasks, whereas others are focused on developing a more rigorous understanding of the underlying mathematical concepts

CSE 258 vs. CSE 158 (vs MGTA, DSC, etc.)

Only one set of lectures, gradescope, Piazza, etc.

(last year's links)

CSE158 and CSE258:

<https://cseweb.ucsd.edu/classes/fa24/cse258-b/>

Lectures

In Lectures I try to cover:

- The basic material (obviously)
 - **Motivation** for the models
 - **Derivations** of the models
- Code examples
- Difficult homework problems / problems from past classes/homeworks/exams etc.

Web Mining and Recommender Systems

Course outline

Course webpage

The course webpage is available here:
<https://cseweb.ucsd.edu/classes/fa25/cse258-a/>

This page will include data, code, slides,
homework and assignments

Course outline

This course is in two parts:

1. Fundamentals:

- Regression
- Classification
- Recommender systems

2. More advanced:

- Content and Structure in Recommendation
- Temporal Data Mining and Recommendation
- Text mining
- (a little bit of visual recommendation?)
- Ethics of Personalized Recommendation

1: Regression

- Linear regression and least-squares
 - (a little bit of) feature design
 - Overfitting and regularization
 - Gradient descent
- Training, validation, and testing
 - Model selection

1. Regression

Product Details

Genres	Science Fiction, Action, Horror
Director	David Twohy
Starring	Vin Diesel, Radha Mitchell
Supporting actors	Cole Hauser, Keith David, Lewis Fitz-Gerald, Claudia Black, Rhiana Gr Angela Moore, Peter Chiang, Ken Twohy
Studio	NBC Universal
MPAA rating	R (Restricted)
Captions and subtitles	English Details ▾
Rental rights	24 hour viewing period. Details ▾
Purchase rights	Stream instantly and download to 2 locations Details ▾
Format	Amazon Instant Video (streaming online video and digital download)

A. Phillips
Reviewer ranking: #17,230,554

90% helpful
votes received on reviews
(151 of 167)

ABOUT ME
Enjoy the reviews...

ACTIVITIES
Reviews (16)
Public Wish List (2)
Listmania Lists (2)
Tagged Items (1)

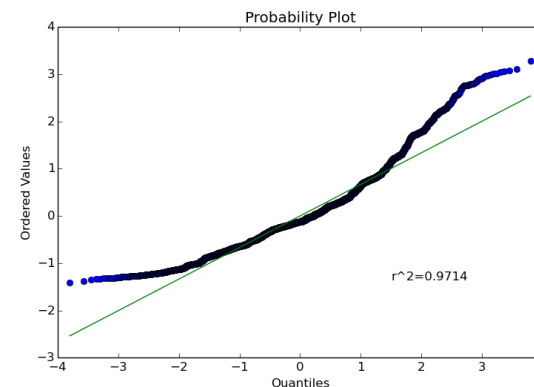


HipCzech
Aficionado
Male, from Texas
Profile Page

Member Since:	Jul 12, 2014	HipCzech was last seen:
Points:	175	Today at 12:19 AM
Beers:	108	
Places:	6	
Posts:	smoother than all of	0
Likes Received:	0	
Trading:	0% 0	

How can we use **features** such as product properties and user demographics to make predictions about **real-valued** outcomes (e.g. star ratings)?

How can we prevent our models from **overfitting** by favouring simpler models over more complex ones?



How can we assess our decision to optimize a particular error measure, like the MSE?

2: Classification

- Logistic regression
- Multiclass and multilabel classification
- How to evaluate classifiers, especially in “non-standard” settings

2: Classification

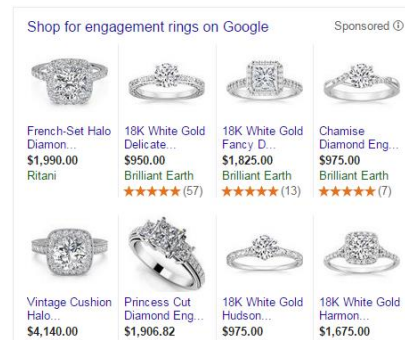
Next we adapted these ideas to **binary** or **multiclass** outputs



What animal is in this image?



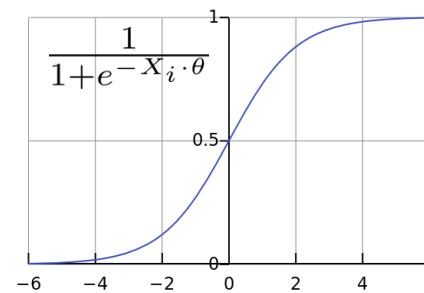
Will I **purchase** this product?



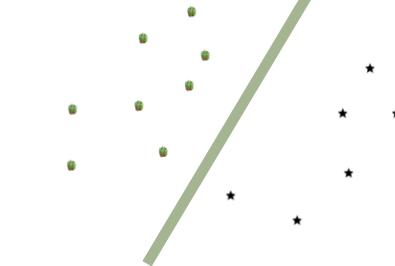
Will I **click on** this ad?



Combining features using naïve Bayes models



Logistic regression

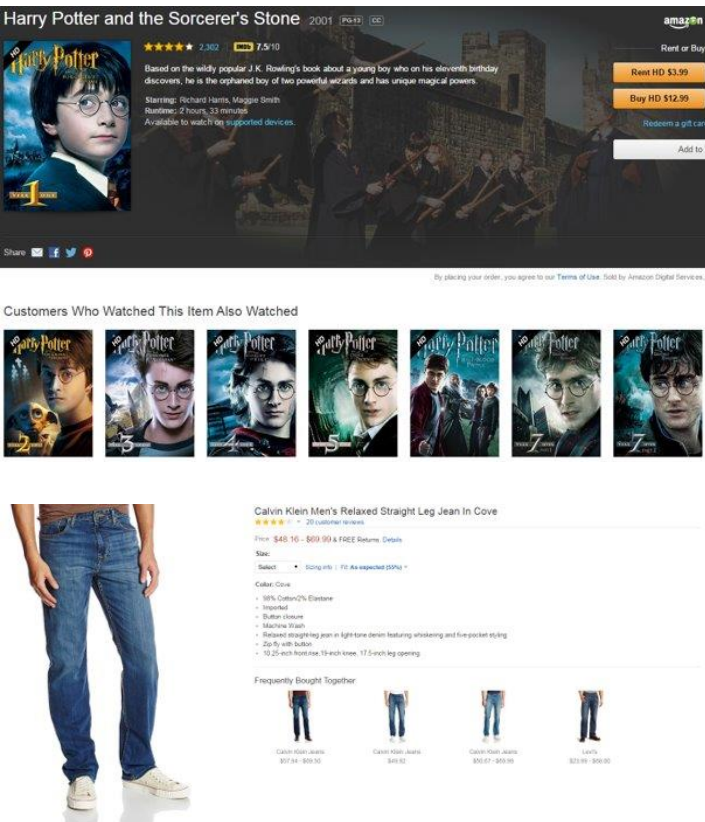


Classifier Evaluation

3: Recommender Systems

- Collaborative filtering (e.g. "people who bought X also bought Y")
- Latent factor models and matrix factorization (e.g. to predict star-ratings)
- Recommender Systems Evaluation

3: Recommender Systems



Harry Potter and the Sorcerer's Stone 2001 PG-13

Based on the wildly popular J.K. Rowling's book about a young boy who on his eleventh birthday discovers he is the orphaned boy of two powerful wizards and has unique magical powers.

Starring: Richard Harris, Maggie Smith
Runtime: 2 hours, 33 minutes
Available to watch on supported devices.

Customers Who Watched This Item Also Watched

Calvin Klein Men's Relaxed Straight Leg Jean In Cove

Price: \$48.10 - \$69.99 & FREE Returns

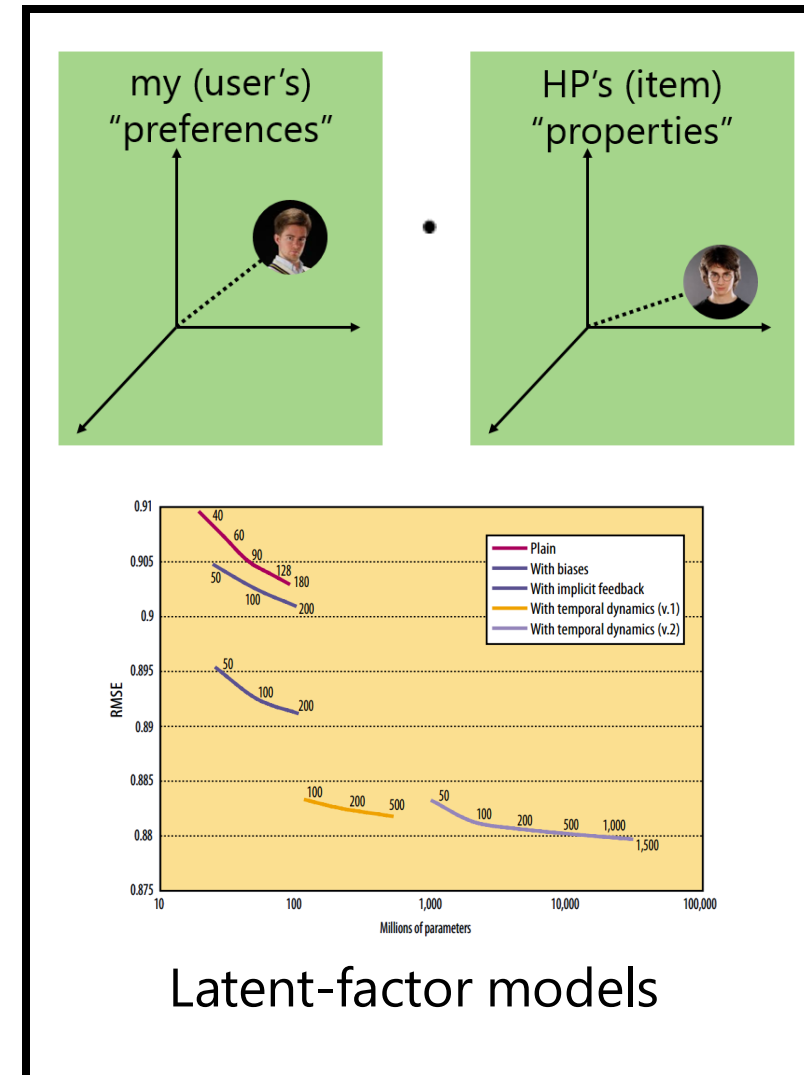
Size: Select: 30x32 (17) 34x32 (20) 36x32 (20) 38x32 (20) 40x32 (20)

Color: Cove

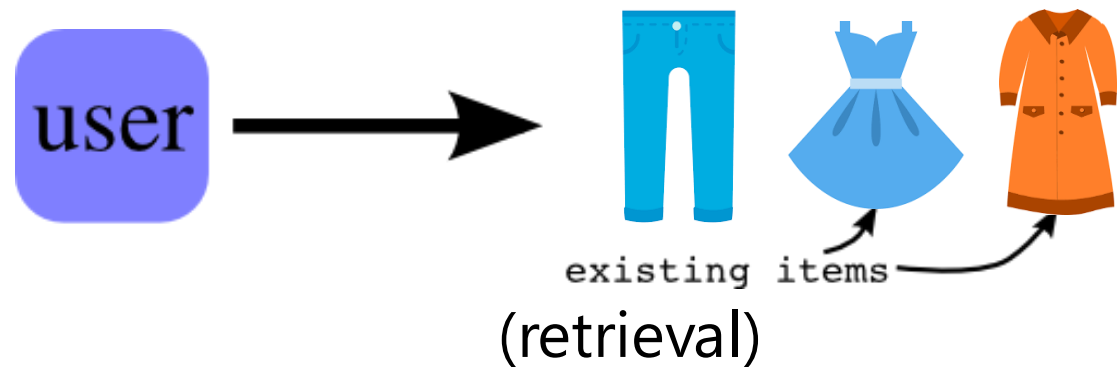
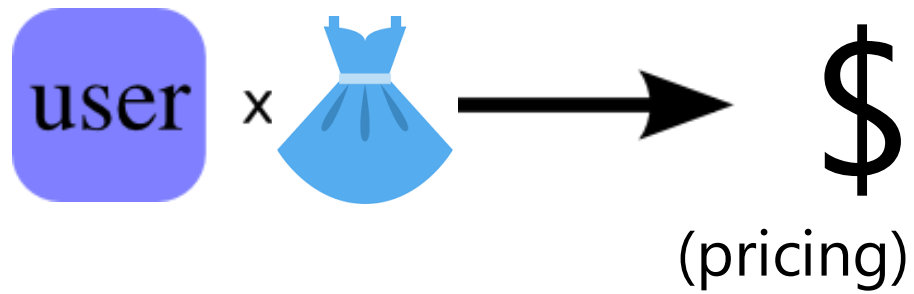
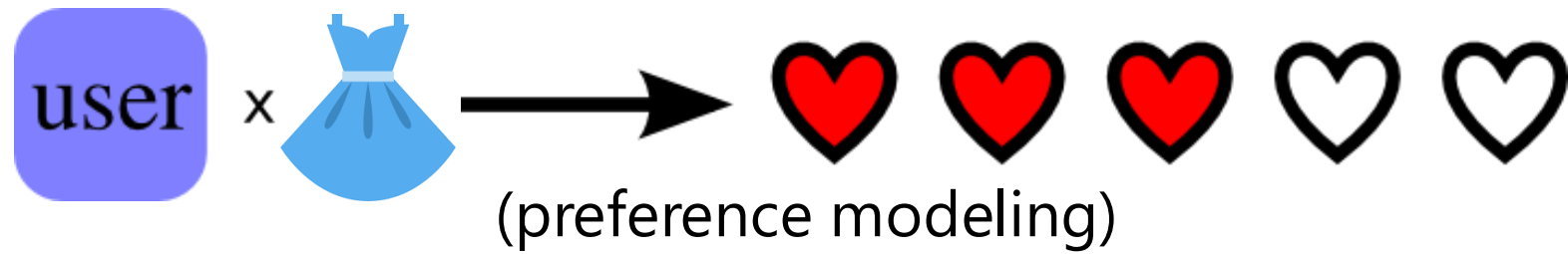
- 100% Cotton/2% Elastane
- Imported
- Shrink Resistant
- Machine Wash
- Relaxed straight leg jean in light tone denim featuring whiskering and five-pocket styling
- Zip fly with button
- 10 25-inch front rise, 19-inch knee, 17.5-inch leg opening

Frequently Bought Together

Similar item recommendation



3: Recommender Systems



4: Content and Structure in Recommender Systems

- Recommendation with complex features
- Recommendation in non-standard settings (e.g. online dating, dynamic pricing, social networks)
 - Advertising
- (maybe discuss some tools/libraries? We'll see how we go for time)

Tools and libraries

- Crawling and parsing data from the Web
- Plotting data
- Tensorflow

```
In [1]: from urllib.request import urlopen
```

```
In [2]: f = urlopen("https://www.goodreads.com/book/show/4671.The_Great_Gatsby")
```

```
In [3]: html = str(f.read())
```

Note: acts like a file object once opened

Note: url of "The Great Gatsby" reviews

```
In [4]: html
```

```
Out[4]: 'b'<!DOCTYPE html>\n<html class="desktop">\n<head prefix="og: http://ogp.me/ns# fb: http://ogp.me/ns/f
b# good_reads: http://ogp.me/ns/fb/good_reads#">\n  <title>\nThe Great Gatsby by F. Scott Fitzgerald\n</title>
\n\n\n  <script type="text/javascript"> var ue_t0=window.ue_t0||+new Date();\n</script>\n  <script type="t
ext/javascript">\n    (function(e){var c=e;var a=c.ue||{
ate();a.d=g;function g(h){return +new Date()-(h?0:a.t0)}f
t:a.d()}}function b(m,l,h,j,i){var k={m:m,f:l,l:h,c:""+
urn false}b.skipTrace=1;e.onerror=b;function f(){c.ue("l
lse)}else{if(e.attachEvent){e.attachEvent("onload",f)}}a.
c.ue=a;c.ueLogError=d("err");c.ues=d("ues");c.uet=d("uet"
a=e.ue||{};function c(g){if(!g){return}var f=d.head||d.g
ateElement("script");h.async="async";h.src=g;f.insertBefo
ages-amazon.com",g=e.ue_cdns||"images-na.ssl-images-amazo
```



Nataliya rated it ★★★★★
childhood-revisited

May 02, 2010

Oh Gatsby, you old sport, you poor semi-delusionally hopeful dreamer with 'some heightened sensitivity to the promises of life', focusing your whole self and soul on that elusive money-colored green light - a dream that shatters just when you are *this* close to it.



Later: More "specialized" material

During the second half of the course I
mostly cover specific topics in some
detail – following are what's planned

Week 6: Midterm (Monday, Nov 4)!

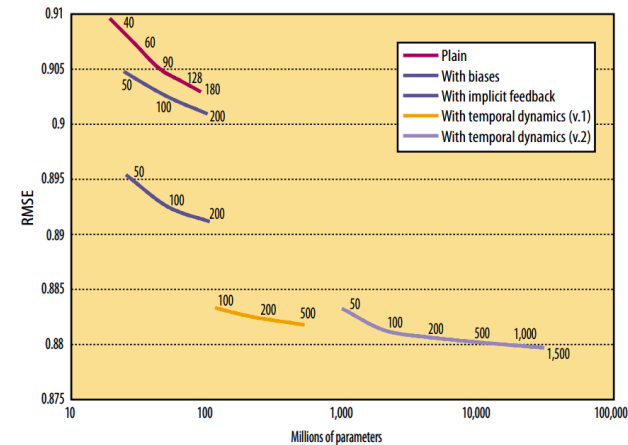
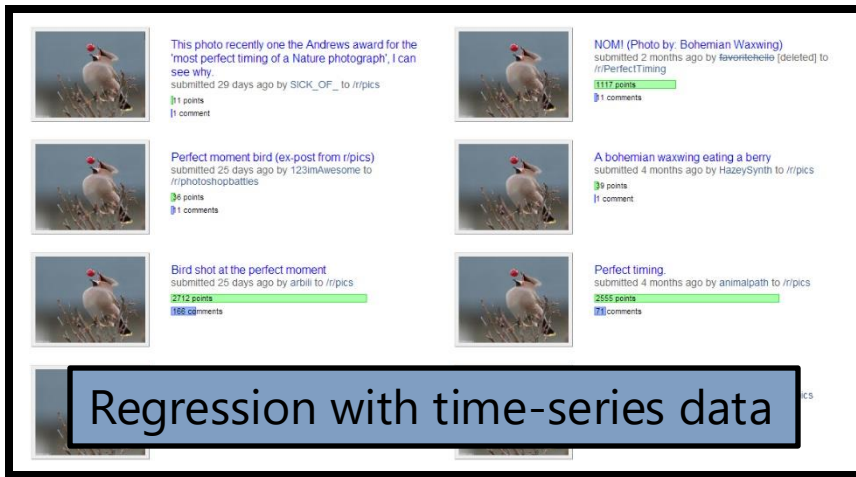
~~Take-home (exact format TBD, not
synchronous but less than 24hr)
(More about grading etc. later)~~

No midterm this year...

5: Temporal & Sequence Data

- Sliding windows & autoregression
 - Temporal dynamics in recommender systems
- Temporal dynamics in text & social networks

6: Temporal & Sequence Data



Temporal dynamics of opinions and preferences (temporal recommendation)

7: Text Mining

- Sentiment analysis
- Bag-of-words representations
 - TF-IDF
 - Stopwords, stemming
- Personalized text models and natural-language explanations

7: Text Mining

yeast and minimal red body thick light a Flavor
sugar strong quad. grape over is molasses lace
the low and caramel fruit Minimal start and
toffee. dark plum, dark brown Actually, alcohol
Dark oak, nice vanilla, has brown of a with
presence. light carbonation. breadly from
retention. with finish. with and this and plum
and head, fruit, low a Excellent raisin aroma
Medium tan

Bags-of-Words

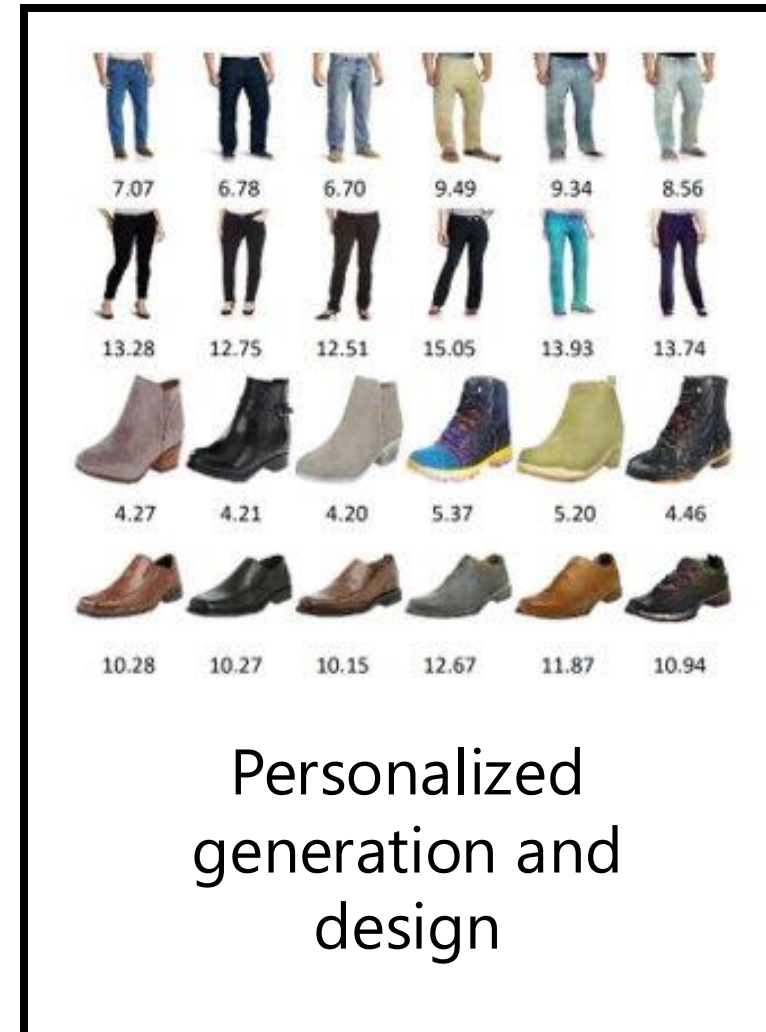
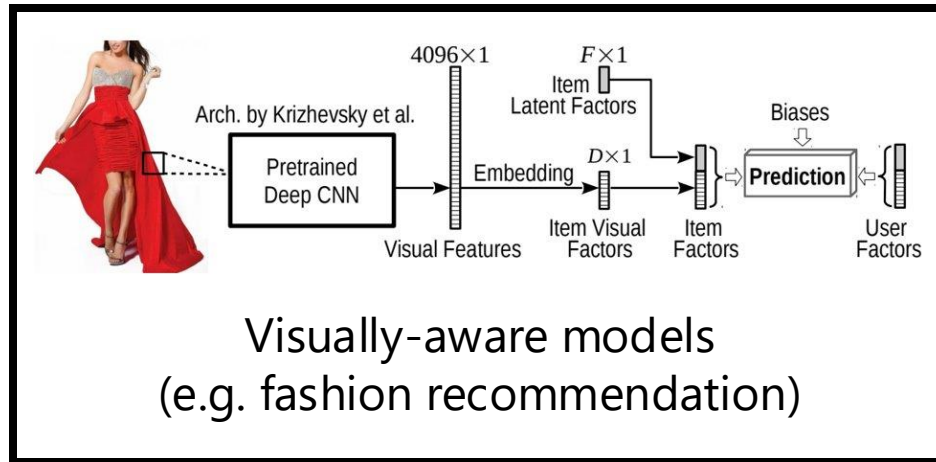
Name: Pomberytini;

Ingredients: pomegranate-blueberry juice,
cranberry juice, vodka;

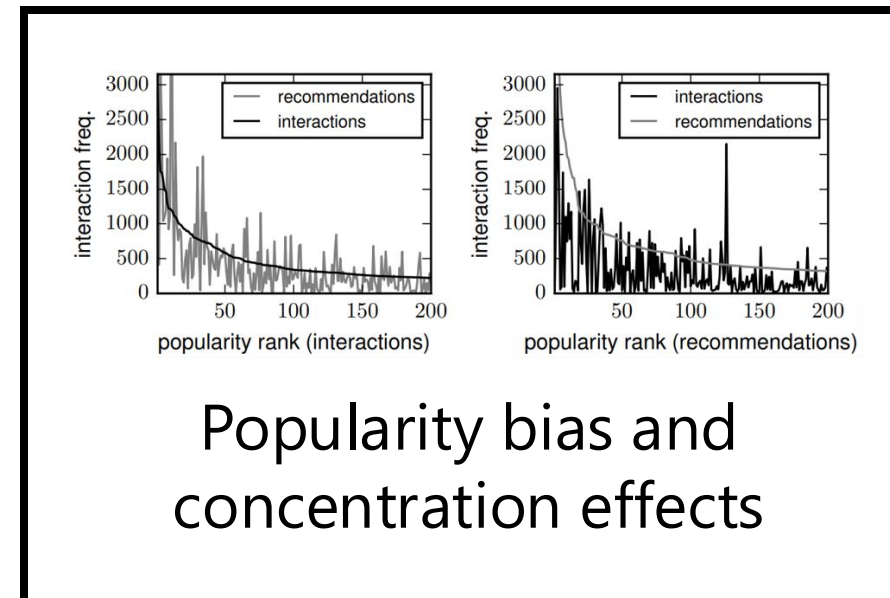
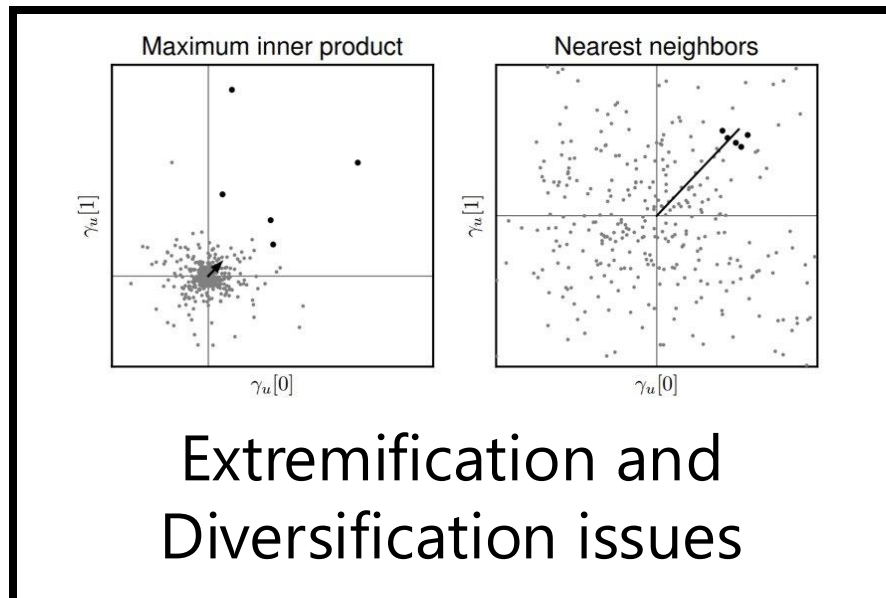
Combine all ingredients except for the ice in a blender or food processor. Process to make a smooth paste and then add the remaining vodka and blend until smooth. Pour into a chilled glass and garnish with a little lemon and fresh mint.



8: Personalized Visual Models



9: Ethics



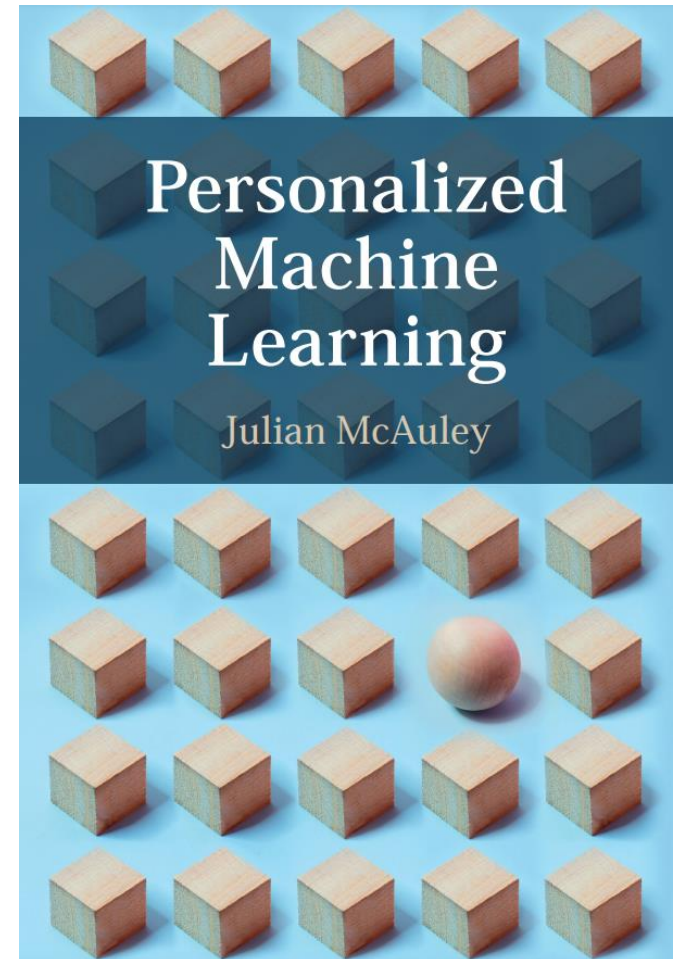
Reading: Textbook

Each lecture corresponds to a chapter from the course textbook:

```
@book{mcauley2022,  
  title = "Personalized Machine Learning",  
  author = "McAuley, Julian",  
  year = "in press",  
  publisher = "Cambridge University Press"  
}
```

All content (including code examples etc.) is available on:

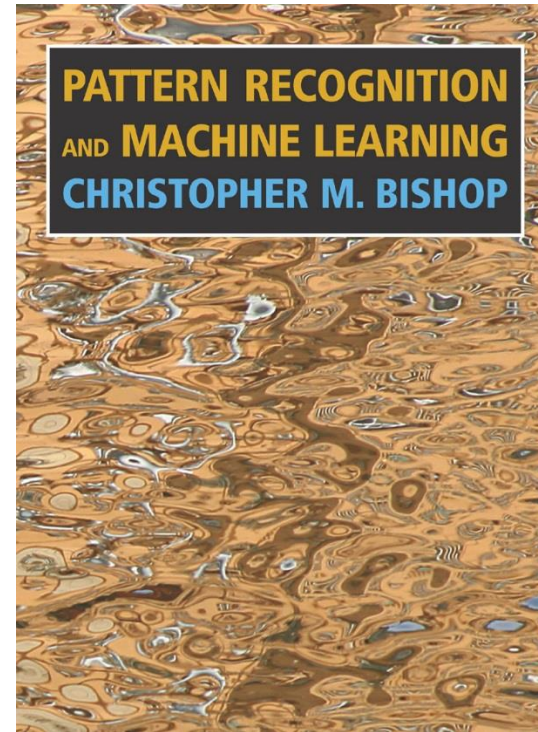
<https://cseweb.ucsd.edu/~jmcauley/pml/>



Reading

I will also give some references from:

- *Bishop: Pattern Recognition and Machine Learning*
- Charles Elkan's notes (http://cseweb.ucsd.edu/classes/fa18/cse258-a/files/elkan_dm.pdf)



Reading

Python Data Products (Coursera)

(this is now
somewhat hidden on
the course webpage)

	Course launched on Apr 3, 2019 Basic Data Processing and Visualization Instructors: Julian McAuley, Ilkay Altintas Launched	Total Enrollments 3524	Active Learners 252	Completion Rate 2.7%
	Course launched on Jul 14, 2019 Deploying Machine Learning Models Instructors: Ilkay Altintas, Julian McAuley Launched	Total Enrollments 1062	Active Learners 117	Completion Rate 1.1%
	Course launched on Apr 3, 2019 Design Thinking and Predictive Analytics for Data Products Instructors: Julian McAuley, Ilkay Altintas Launched	Total Enrollments 1575	Active Learners 76	Completion Rate 2.6%
	Course launched on Apr 17, 2019 Meaningful Predictive Modeling Instructors: Julian McAuley, Ilkay Altintas Launched	Total Enrollments 1109	Active Learners 39	Completion Rate 2.1%

Evaluation

- There will be **four** homework assignments worth 8 marks. Your **lowest grade** will be dropped, so that 4 homework assignments = 24 marks
- ~~There will be a (take-home) midterm in week 6, worth 25%~~
- One assignment on recommender systems (after week 5), worth 27 marks
- A short open-ended assignment, worth 25 marks
- 4 marks for peer grading Assignment 2

Evaluation

HW = 24 marks

Assignment 1 = 27

Assignment 2 = 25

Peer grading = 4

Actual goals:

- Understand the basics and get comfortable working with data and tools (HW)
- Build something that **actually works** (Assignment 1)
- Apply your knowledge creatively (Assignment 2)

Evaluation

- Homework should be delivered by **Monday night**
- All submissions will be made **electronically** (instructions will be in the homework spec, on the class webpage)

Evaluation

Schedule (subject to change but hopefully not):

Week 1: Hw 1 out

Week 3: Hw 1 **due**, Hw2 out

Week 5: Hw 2 **due**, Hw3 out, Assign. 1 out

Week 7: Hw 3 **due**, Hw4 out, Assign. 2 out

Week 8: Assignment 1 **due**

Week 9: Hw4 **due**

Week 10: Assignment 2 **due**

Evaluation

All homework is to be submitted on
gradescope

- Gradescope links will be posted to Piazza
 - Make sure you submit to the correct gradescope! (MGTA461 students should submit to 258)
- Will continue **autograding** homeworks this year

Academic integrity

- Autograding and standardized code submission will (hopefully) make it easier to detect plagiarized submissions
- Cases of cheating in previous years had quite serious consequences...

Grade inflation!

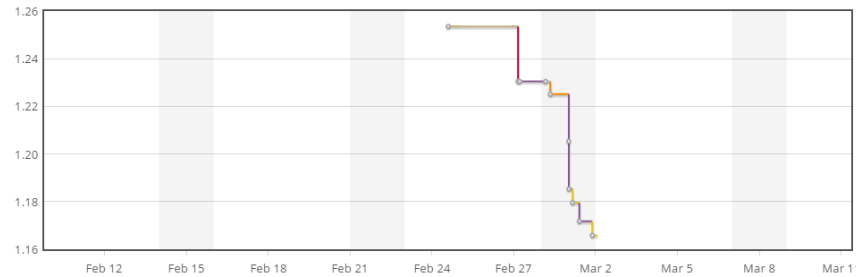
- Between pandemics, strikes, peer-grading, autograding, etc., grades have increased somewhat... but I'm trying to keep it under control
- Don't believe your friends who tell you it's the easiest class ever!

Web Mining and Recommender Systems

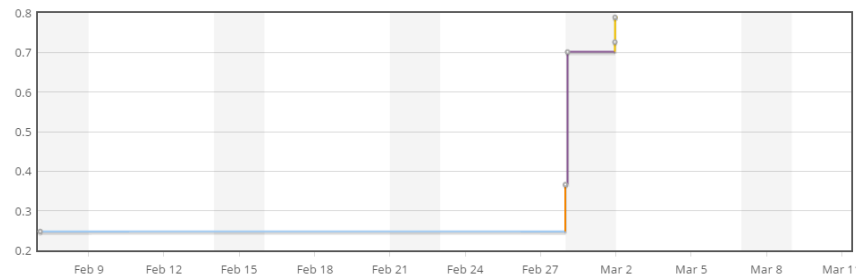
Some Previous Assignments

Assignment 1

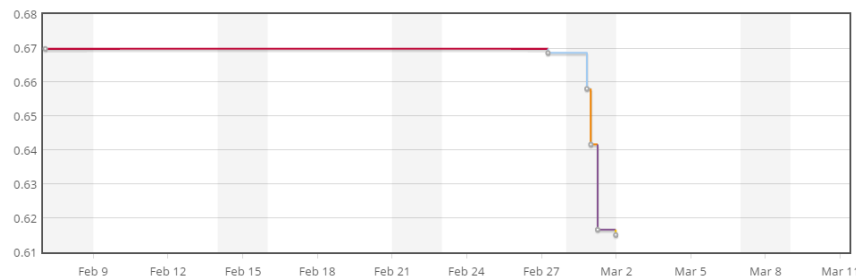
- Prediction tasks on (e.g.) Amazon clothing data, run as a competition on gradescope



Rating prediction



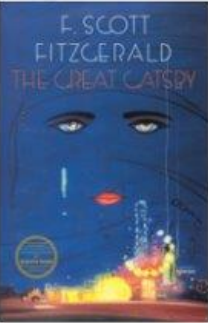
Purchase prediction



Helpfulness prediction

Assignment 1

- We'll do something similar this year, but on Goodreads (probably?) data



The Great Gatsby
by F. Scott Fitzgerald. Jake Gyllenhaal (Narrator)

★★★★☆ 3.9 · [Rating details](#) · 3,090,834 Ratings · 56,204 Reviews

A true classic of twentieth-century literature, this edition has been updated by Fitzgerald scholar James L.W. West III to include the author's final revisions and features a note on the composition and text, a personal foreword by Fitzgerald's granddaughter, Eleanor Lanahan—and a new introduction by two-time National Book Award winner Jesmyn Ward.

[Want to Read](#) ▼

Rate this book
★★★★☆

[Open Preview](#)

[Kindle Store \\$12.99](#) [Amazon](#) [Stores ▼](#) [Libraries](#)

Paperback, US / CAN, 180 pages
Published September 2004 by Scribner (first published April 1925)

[More Details...](#) [edit details](#)



Nataliya rated it ★★★★★ [May 02, 2010](#)
[Show Nataliya's Goodreads bookshelves](#), [2013-reads](#), [i-also-saw-the-film](#), [books-from-childhood-revisited](#)

Oh Gatsby, you old sport, you poor semi-delusionally hopeful dreamer with 'some heightened sensitivity to the promises of life', focusing your whole self and soul on that elusive money-colored green light – a dream that shatters just when you are *this* close to it.



Jay Gatsby, who dreamed a dream with the passion and courage few possess – and the tragedy was that it was a wrong dream colliding with reality that was even more wrong – and deadly.

Just like the Great Houdini – the association the [...more](#)

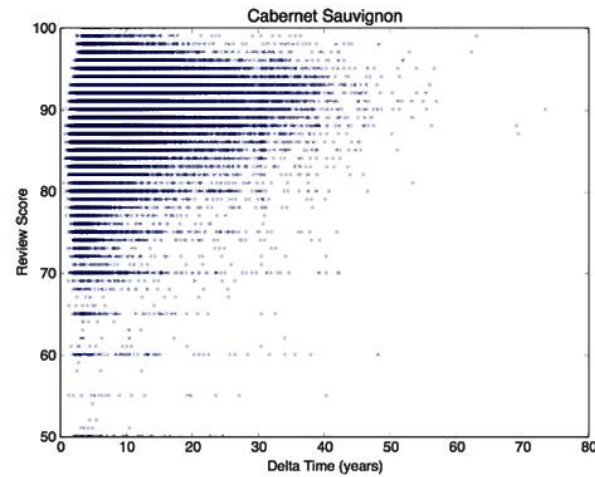
713 likes · [Like](#) · [see review](#)



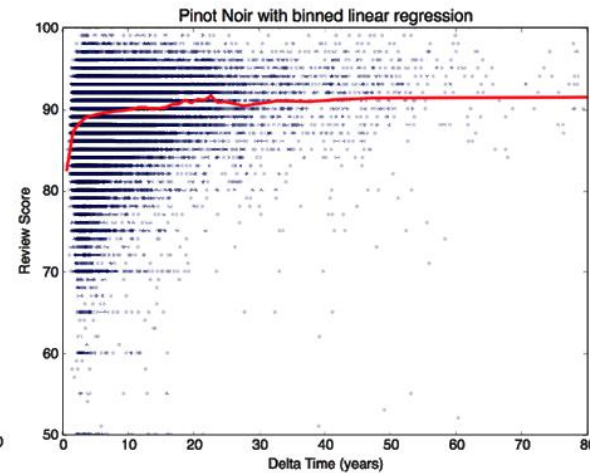
Alex rated it ★★★★★ [Dec 24, 2007](#)

The Great Gatsby is your neighbor you're best friends with until you find out he's a drug dealer. It charms you with some of the most elegant English prose ever published, making it difficult to discuss the novel without the urge to stammer awestruck about its beauty. It would be evidence enough to

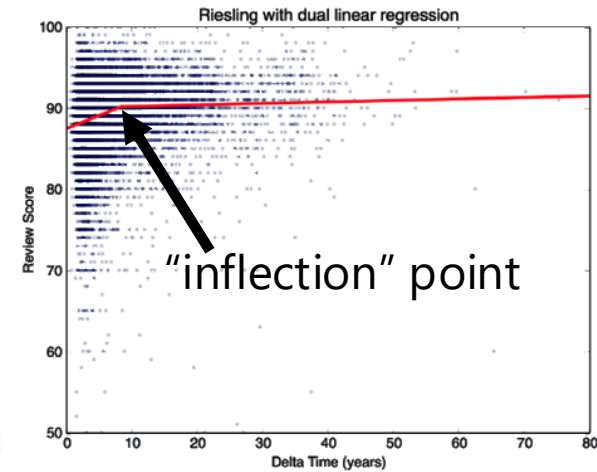
Assignment 2



Raw rating data

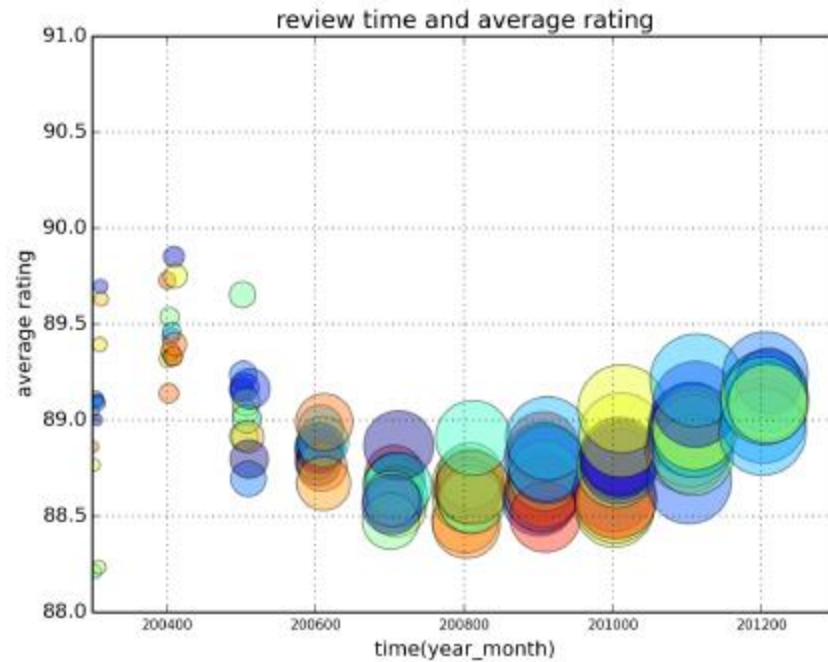


binned regression

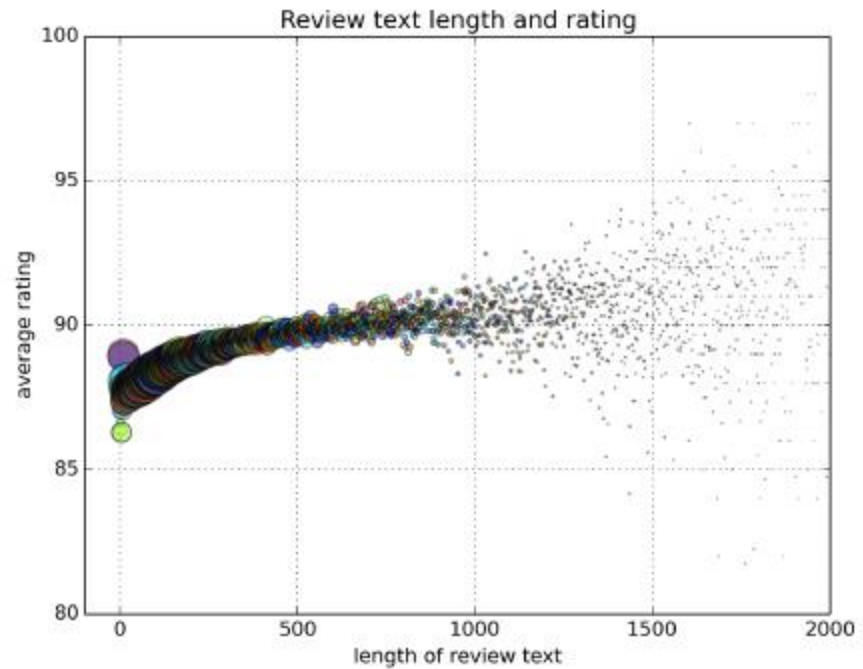


dual regression

Assignment 2



ratings vs. time

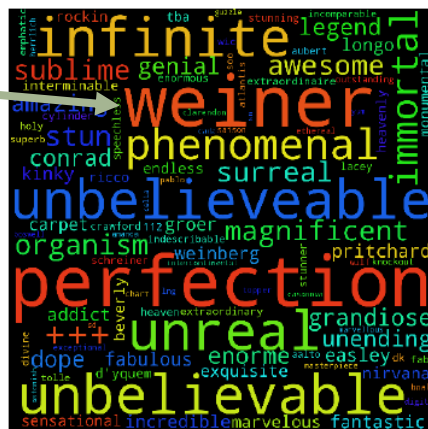


ratings vs. review length

Assignment 2

?

cellartracker:

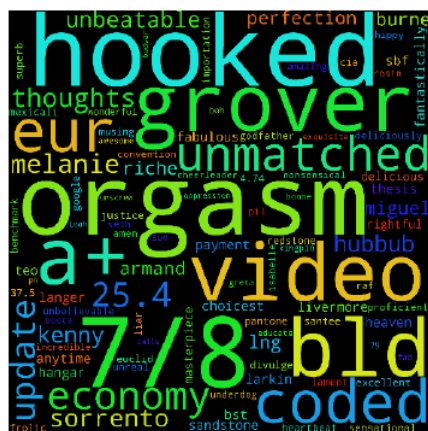


positive words in wine reviews

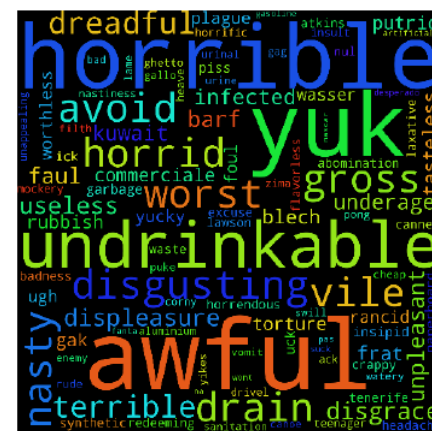


negative words in wine reviews

RateBeer:



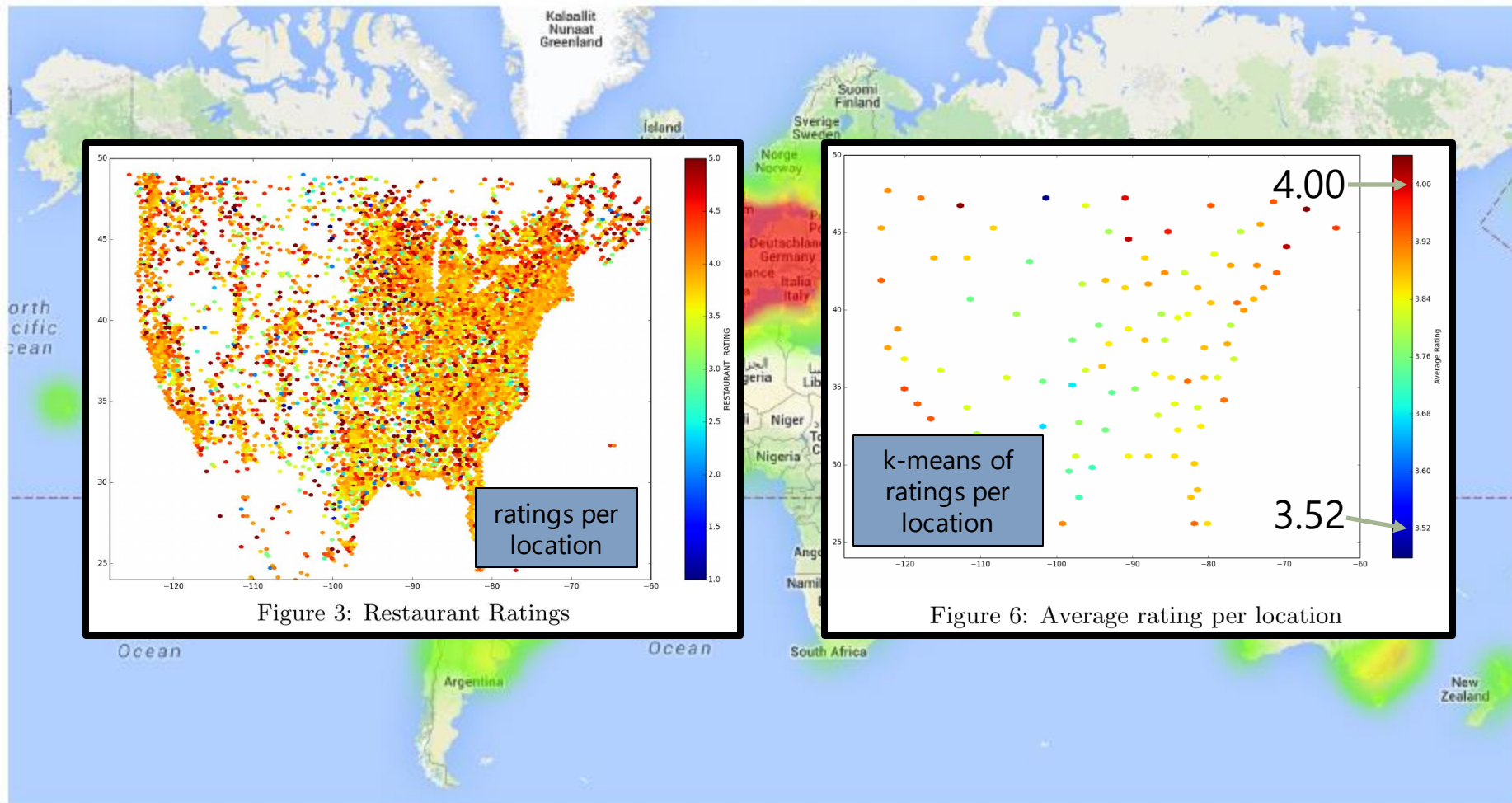
positive words in beer reviews



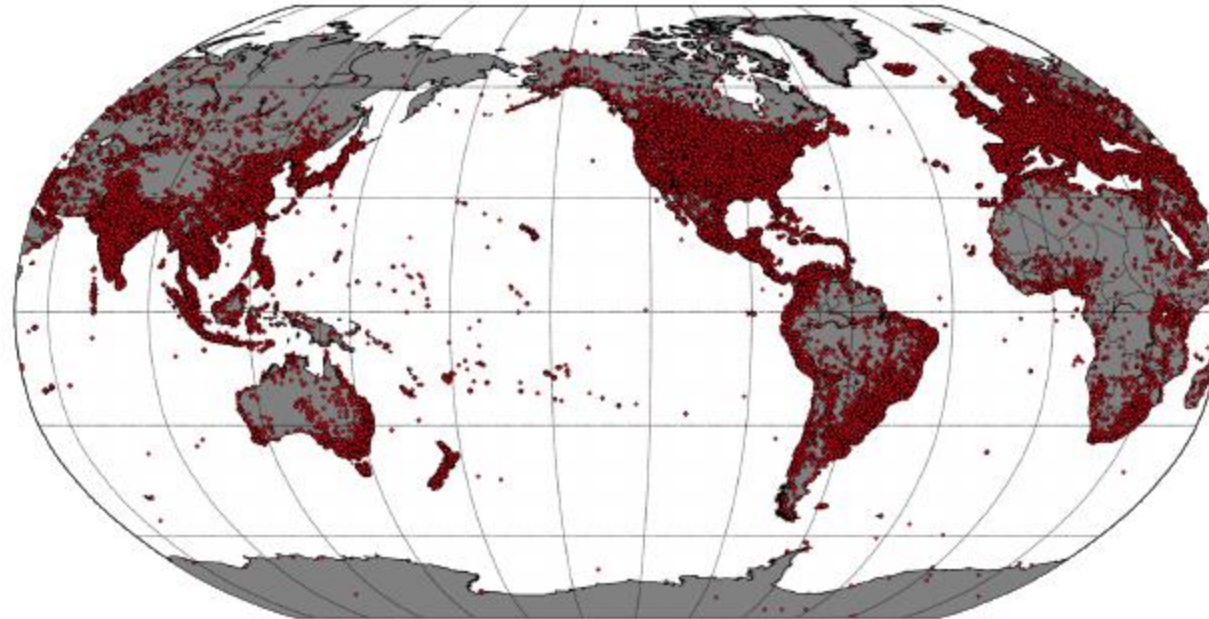
negative words in wine reviews

Ben Braun & Robert Timpe – “Text-based rating predictions from beer and wine reviews”

Assignment 2



Assignment 2

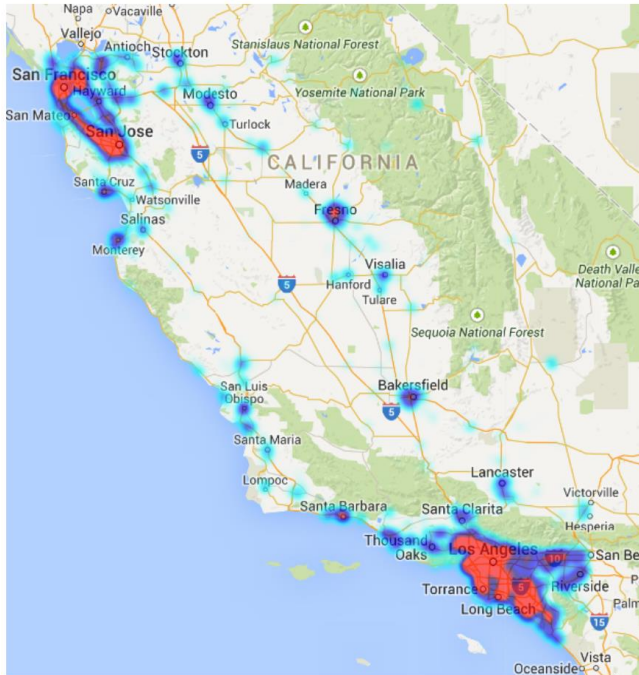


$$\widehat{r_{ui}} = \mu + b_u + b_i + (q_i + \frac{1}{|M(i)|} \sum_{n \in M(i)} |s_n|)^T p_u$$

set of geographic neighbours

impact of neighbours

Assignment 2



<i>"Fitness"</i>	<i>"Italian Restaurants"</i>	<i>"Airport & Rentals"</i>	<i>"Computer Repairs"</i>	<i>"Mexican"</i>
gym	food	san	computer	food
training	restaurant	francisco	store	mexican
fitness	wine	car	phone	tacos
classes	menu	airport	system	burrito
equipment	great	jose	buy	good
class	delicious	time	laptop	salsa
life	service	rental	apple	taco
great	dinner	driver	repair	chips
workout	dishes	service	problem	burritos
weight	excellent	bus	back	fish
ve	dining	shuttle	fixed	chicken
work	meal	taxi	pc	place
body	italian	trip	drive	delicious
yoga	experience	city	price	love
trainers	amazing	cab	data	fresh
people	wonderful	lax	fix	great
years	atmosphere	area	iphone	beans
feel	small	experience	screen	restaurant
instructors	decor	company	bought	asada

Topic model from Google Local business reviews

Assignment 2

Wikispeedia
navigation
traces:

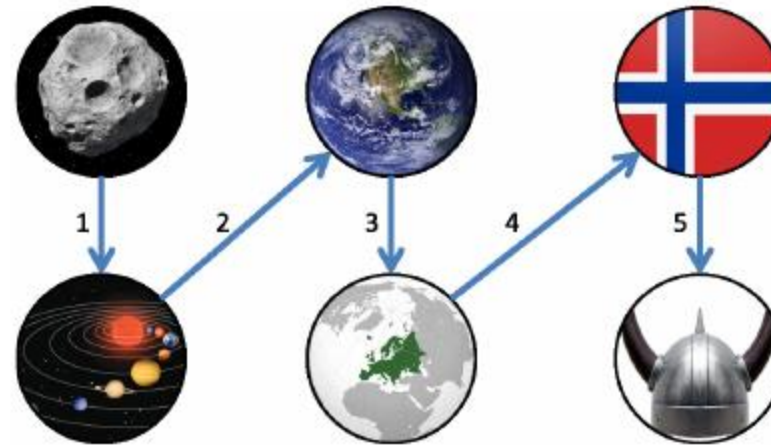


Figure 5: Graph of a complete path

	Average Click	Average Time
Finish Path	4.72	158.27
Finished Path Back	6.75	158.31
Unfinished Path	2.97	835.29
Unfinished Path Back	5.2	836.00

Assignment 2

Images from Chictopia →



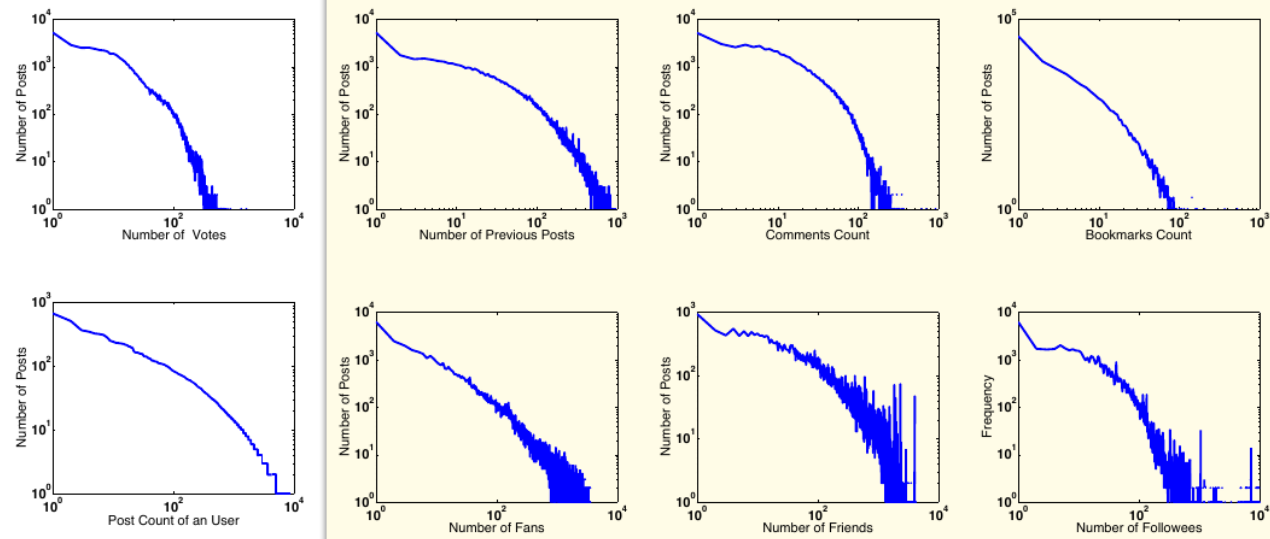
Tags electric, every day, summer, cute, T-shirt, chic

Clothes Chartreuse Uniqlo Socks
Light Blue Uniqlo T-Shirt
Bubble Gum Tie-Ups Belt
White Christian Louboutin Heels

User Information 1369 friends
15 followees
2245 fans

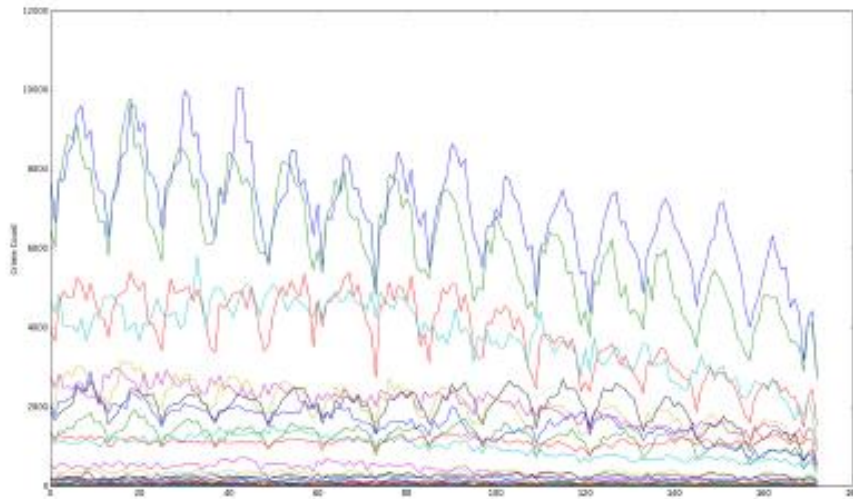
Popularity 129 votes
62 comments
15 bookmarks

Power laws! →

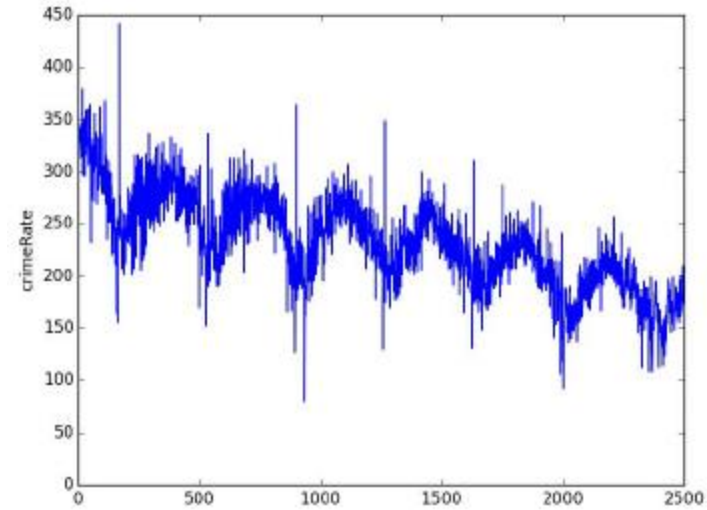


Wei-Tang Liao & Jong-Chyi Su – “Image Popularity Prediction on Social Networks”

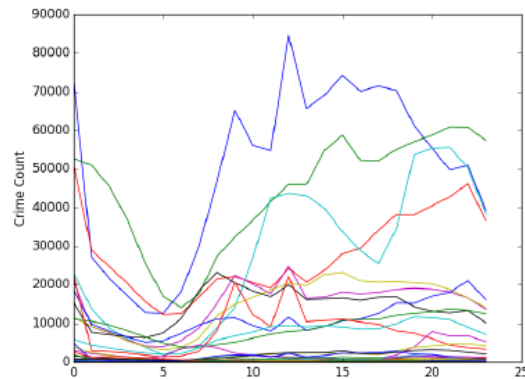
Crime (Chicago)



Over 15 years



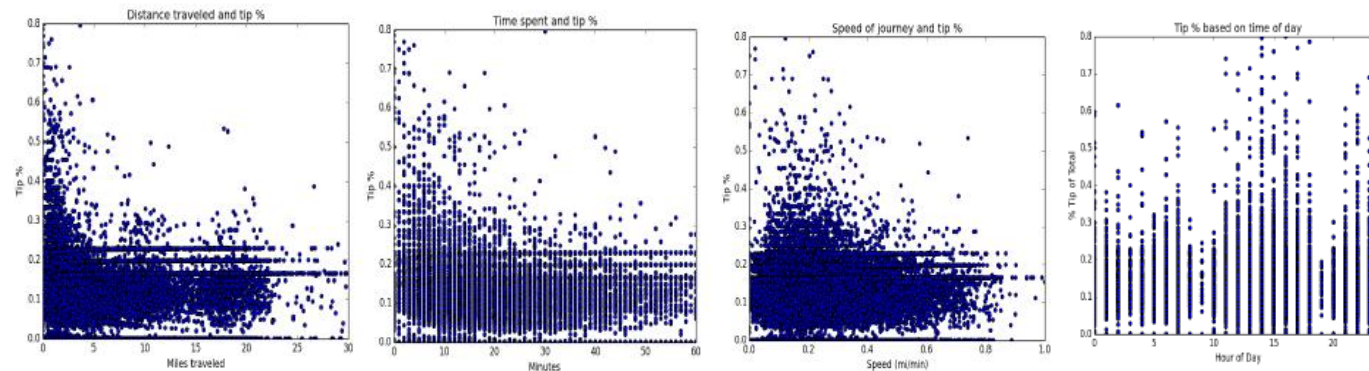
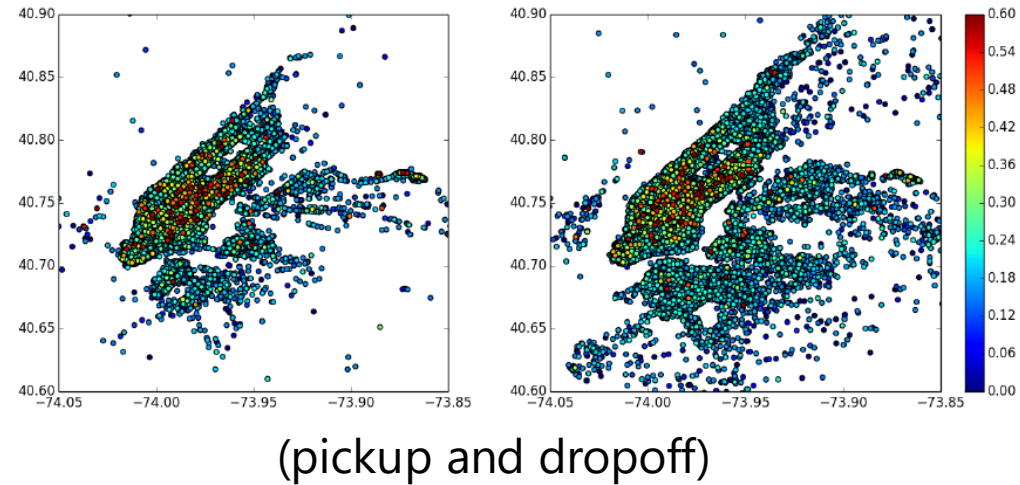
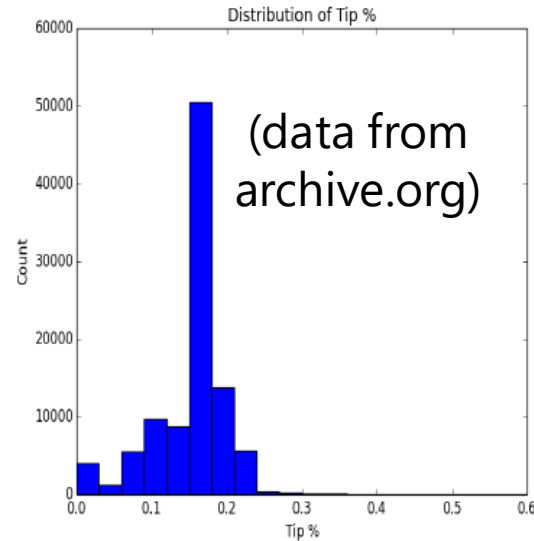
Over 7 years



Hour of the day

Goal: to predict the number of incidents of crime on a given day

Predicting Taxi Tip-Rates in NYC



Distance, time taken, speed, and time of day (also on geo)

TAs

TAs will introduce themselves on Piazza, and advertise their office hours there

Office hours

- Office hours (TAs and mine) will be held throughout the week (exact times and links will be posted on Piazza)

Thanks!

Most announcements will be
posted to Piazza

<https://piazza.com/ucsd/fall2024/cse158/home/>
<https://piazza.com/ucsd/fall2024/cse258/home/>

please participate!

Please sign up for my other classes!

- Fairness & interpretability
W12S → 190/293
- ML for music
Sp25 → 153/253